

Use case: Upgrading Windows

You can use Tanium Deploy to prepare and deploy Windows 10 Upgrades to your Windows endpoints. To complete a Windows 10 Upgrade, you must understand how to target and deploy software packages in Deploy.

Overview

Deploy supports the upgrade of Windows 7 Service Pack 1 and later to Windows 10 and upgrades of Windows 10 to newer builds of Windows. You can use Deploy to handle the Windows upgrade process in three phases:

Phase 1: Download Windows media and scan for compatibility

The Phase 1 software package invokes a PowerShell script to copy the files that are required for the upgrade to the `C:\Deploy` directory on the endpoint. After the files are in place, the PowerShell script runs Windows Setup to check for upgrade compatibility. The script then takes the information returned from Windows Setup and records the results in the registry.

Phase 2: Compatibility remediation and re-scan

If the Phase 1 compatibility scan fails, you must remediate any problems. You can use the Phase 2 package to force a new scan. Rescanning is necessary if there is a compatibility problem detected in the Phase 1 scan that is later remediated.

Phase 3: Windows Upgrade

The Phase 3 package invokes a PowerShell script that runs Windows Setup to perform the upgrade.

Before you begin

To set up your Windows 10 upgrade, you must have the Windows 10 media (ISO or ESD file) that corresponds to the Windows 10 Windows version, channel, architecture, and language that you want to deploy.

Your security administrator must create security exclusions to ensure successful operation of Tanium Deploy and the Tanium Client. Additional security exclusions for Windows are required for Windows upgrades.

- [Microsoft Support: Virus scanning recommendations for Enterprise computers that are running currently supported versions of Windows \(KB822158\)](#).
- [Tanium Client Management User Guide: Host system security exclusions](#)
- [Security exclusions on page 30](#)



NOTE

If you have any questions about implementing Windows upgrades in your environment, [Contact Tanium Support on page 75](#) to discuss your testing and implementation plans.

Import software packages

To begin, you must import three software packages from the predefined package gallery. The following examples use Windows 10 Version 21H1, 64-bit. Substitute with the version and architecture you are deploying as needed.

1. From the Deploy menu, go to **Software**, and then click **Predefined Package Gallery**.



You can use the filter or search options to narrow the list to Microsoft Windows upgrades.

2. Select the following three packages:
 - **InPlace Upgrade to Windows 10 Version 21H1 x64 - Phase1 - Pre-Cache**
 - **InPlace Upgrade to Windows 10 Version 21H1 x64 - Phase2 - Pre-Scan**
 - **InPlace Upgrade to Windows 10 Version 21H1 x64 - Phase3 - Upgrade**
3. Click **Import**, confirm the action, and then click **Go To Software Packages**.



A warning might appear indicating there are pending software package changes and that the catalog must be distributed. You can skip distributing the catalog in this step, as you need to do it again after providing the Windows 10 media.

Review and modify software packages

Before you deploy the packages, you must upload the downloaded ISO or ESD files, and then make some modifications to the software packages in the following order:

1. [Modify the Phase 1 software package on page 77](#)
2. [View the Phase 2 software package on page 78](#)
3. [\(Optional\) Modify the Phase 3 software package on page 78](#)

Modify the Phase 1 software package

1. From the Deploy menu, go to **Software**.



You can use the filter or search options to narrow the list to Microsoft Windows upgrades.

2. Click **InPlace Upgrade to Windows 10 Version 21H1 x64 - Phase1 - Pre-Cache** and then click **Edit**.
3. Click **Add Package Files** and then click **Local File**.

4. Navigate to the ISO or ESD file and then click **Open**.



This upload takes a while to complete. If you want to complete other tasks in Deploy, click **Save and Finish Later**. The upload continues in the background and you can return to complete the package later.

After the upload completes, the file entry with its SHA-256 hash appears.

5. Click **Update Package** and then if prompted, click **Distribute Catalog**.



After you update the package, the Module Server transfers files to the Tanium Server. This process could take up to 30 minutes. You cannot deploy the Phase 1 package to any clients until it completes.

6. (Optional) If a locale other than en-us is required, edit the **Update Detection** and **Install Verification** rule and change **1033** to the appropriate language ID. For more information and a complete list of all language/region decimal IDs, see [Microsoft Documentation: Available languages for Windows](#).

View the Phase 2 software package

This package triggers a new scan after remediating any problems with the Phase 1 scan. Modifications to this software package are not required or supported.

(Optional) Modify the Phase 3 software package

The PowerShell script associated with Phase 3 starts the Windows Setup upgrade process and in most cases should need no modification. The script runs Windows Setup with the following command line arguments:

```
/auto Upgrade /NoReboot /Quiet /DynamicUpdate disable /ShowOOBE none /Telemetry  
disable /Uninstall enable
```

If you need to append additional arguments to the command line that Windows Setup is using for upgrade, you can complete the following steps:



If you modify these packages, ensure that you test thoroughly and run them at your own risk.

1. From the Deploy menu, go to **Software**.



You can use the filter or search options to narrow the list to Microsoft Windows upgrades.

2. Click **InPlace Upgrade to Windows 10 Version 21H1 x64 - Phase3 - Upgrade** and then click **Edit**.
3. In the **Update** section of the **Deploy Operations** section, add any arguments to the end of the command in **Run Command**. For example, if you want to add the option to keep BitLocker active during a Windows 10 to Windows 10 upgrade, you can append the following command line:

```
powershell.exe -executionpolicy remotesigned -noninteractive -command  
".\Invoke-Win10Upgrade.ps1" /Bitlocker ForceKeepActive
```



The full list of commands available to Windows Setup can be found at: [Microsoft Documentation: Windows Setup Command-Line Options](#).

Deploy software packages

After the software packages are ready, complete the following steps to complete the upgrade process:

1. [Deploy the Phase 1 package on page 79](#)
2. [Review and remediate compatibility results on page 79](#)
3. [Deploy Tanium package: registry - set value on page 81](#)
4. [Deploy the Phase 2 package on page 82](#)
5. [Deploy the Phase 3 package on page 82](#)

Deploy the Phase 1 package

You cannot complete the following steps until the Windows 10 media you uploaded in [Modify the Phase 1 software package on page 77](#) has finished caching on the Tanium Server.

1. From the Deploy menu, go to **Software**.



You can use the filter or search options to narrow the list to Microsoft Windows upgrades.

2. Select **InPlace Upgrade to Windows 10 Version 21H1 x64 - Phase1 - Pre-Cache** and click **Deploy Package**.
3. Select the desired options according to your environment, click **Save**, and then click **Deploy Software**.



For this deployment, notifications are not necessary because actions are not visible to end users.



Deploy this package on an ongoing basis. Deploy automatically runs this deployment on endpoints when they enter an eligible state.

Review and remediate compatibility results

Deploy includes sensors that are designed to help you determine endpoint readiness for the Windows 10 upgrade.

Deploy - Windows Upgrade Ready (Optional)

This sensor returns a True/False result as to whether systems are ready and meet the Windows 10 requirements for installation. A False result means that the scan failed for a variety of reasons or that you targeted an ineligible system, such as Windows Server, Mac, or Linux. Use this question for tracking counts of endpoints that are ready for upgrade.

Deploy - Windows Upgrade Scan Details

This sensor provides detailed information from Windows Setup, including specific compatibility blockers. Use this sensor for remediation in the following procedure.

Deploy - Windows Upgrade Scan Results (Optional)

This sensor produces the text of the scan and the result (return code) from Windows Setup. An example of a successful return code from Windows Setup is: 0xc1900210. Use this sensor for tracking groups of compatibility states. Review Windows Upgrade SCAN Details and remediate errors.

Use the **Deploy - Review Windows Upgrade Scan Details** sensor to view more detailed information about compatibility failures. For example, some installed software might be incompatible with the upgrade process to this version of Windows. In this scenario, you might need to update or remove software before upgrading Windows.

You can use the **Deploy - Review Windows Upgrade Scan Results** sensor to review and remediate any conditions that prevent upgrade, such as low disk space or a compatibility block.

1. Go to the Tanium **Home** page and in the **Explore Data** field, ask the following question:

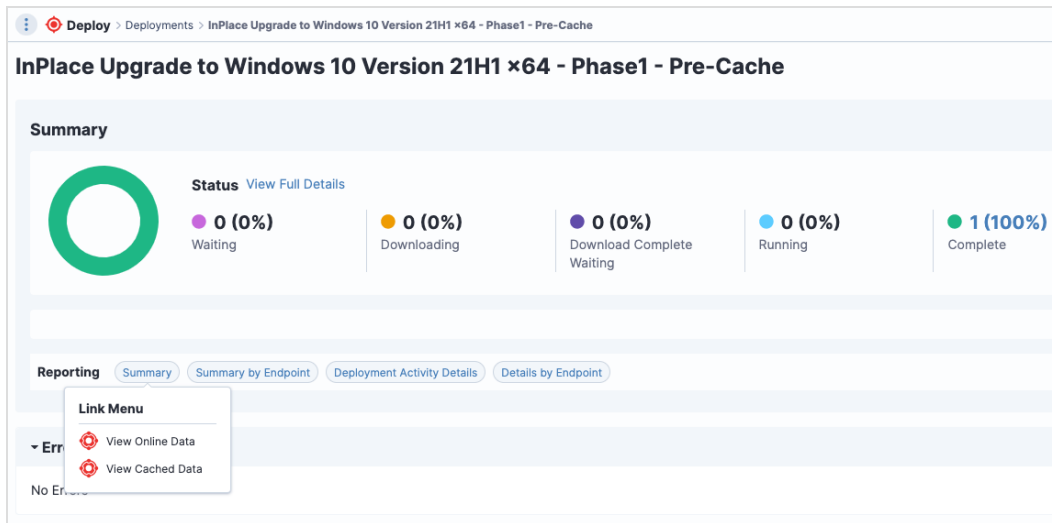
```
Get Windows Upgrade Scan Details from all machines
```

2. (Optional) Filter to restrict the question to endpoints that you targeted with the Phase 1 deployment.
 - a. From the Deploy menu, go to **Deployments** and click the **InPlace Upgrade to Windows 10 Version 21H1 x64 - Phase1 - Pre-Cache** deployment.



You can use the filter or search options to narrow the list to Microsoft Windows upgrades.

- b. In the **Summary** section, click **Summary** next to **Reporting** and select **View Online Data**.



A new Interact page appears with a question filtered to machines that ran this deployment.

- c. Replace the `Get Deploy - Deployments` portion of the question with `Get Deploy - Windows Upgrade Scan Details`.
- d. Select the new question for the filtered view of **Get Deploy - Windows Upgrade Scan Details**.
3. Identify remediation strategies for blocking conditions. For example, if an answer indicates an installed application is not compatible, deploy a Deploy Software Package or Tanium Package to update or remove that application.
4. (Optional) If remediating with a Deploy Software Package or Tanium Package, consider using the **Custom Tagging - Add Tags** Tanium Package to apply a custom tag before deploying the remediation to easily track the remediated endpoints. For more information, see [Tanium Community: How to Group Computers Based on Your Organization's Needs Using Custom Tags](#).

Deploy Tanium package: registry - set value

After you have remediated any blocking issues, you must set the registry value to enable Phase 2 to run.

1. Ask a Tanium question that identifies the remediated endpoints. For example, if you used a custom tag called **Win10remediated** to identify endpoints in the previous step, ask the following question:

```
Get Online from machines with custom tags equals Win10Remediated
```

2. Select the answer indicating the remediated endpoints and click **Deploy Action**.
3. Select the Tanium Package: **Registry - Set Value**.
4. Select the **OS Architecture** according to the targeted endpoints.
5. For **Registry Key Name**, enter `HKEY_LOCAL_MACHINE\Software\WOW6432Node\Tanium\Tanium Client\OSD`.

6. For **Value Name**, enter `Status`.
7. For **Value Data**, enter `WIM File Copied`.
8. For **Value**, select **REG_SZ**.

After you are ready to rescan any endpoints, proceed to [Deploy the Phase 2 package on page 82](#) to force a compatibility re-scan.

Deploy the Phase 2 package

The Phase 2 deployment uses the Windows Setup files to run the same compatibility scan that runs during the Phase 1 deployment. If there were no compatibility errors during the Phase 1 deployment, the Phase 2 deployment is not required and all endpoints that completed the Phase 1 deployment have the **Installed** applicability status. If errors were remediated and the registry value was set, as described in [Deploy the Phase 2 package on page 82](#), the Phase 2 deployment appears in the **Update Eligible** applicability status on those endpoints.

1. From the Deploy menu, go to **Software**.



You can use the filter or search options to narrow the list to Microsoft Windows upgrades.

2. Select **InPlace Upgrade to Windows 10 Version 21H1 x64 - Phase2 - Pre-Scan** and click **Deploy Package**.
3. In the **Targeting** section, select the same targets or a subset of the targets that you configured for the Phase 1 deployment.
4. Select the desired options according to your environment, click **Save**, and then click **Deploy Software**.



For this deployment, notifications are not necessary because you are running a scan that is not visible to end users.



Deploy this package on an ongoing basis. Deploy automatically runs this deployment on endpoints when they enter an eligible state.

After you deploy the Phase 2 package, review compatibility results again for any targeted endpoints that are not in **Installed** status for the Phase 2 package. Some endpoints might take several iterations of remediation before the compatibility scan passes.

Deploy the Phase 3 package

The Phase 3 deployment depends on the successful completion of the Phase 1 and Phase 2 deployments. This deployment executes the Windows Upgrade.

1. From the Deploy menu, go to **Software**.



You can use the filter or search options to narrow the list to Microsoft Windows upgrades.

2. Select **InPlace Upgrade to Windows 10 Version 21H1 x64 - Phase3 - Upgrade**, click **Deploy Package**.

3. In the **Targeting** section, select the same target as the Phase 1 deployment or the desired targets that you want to start the upgrade. Endpoints that are not in **Installed** status for both the Phase 1 and Phase 2 deployments cannot execute this software package operation, so there is no need to restrict targeting based on the previous phase results.
4. Select the desired deployment options based on your environment, click **Save**, and then click **Deploy Software**.



BEST PRACTICE

Pre-notify the user, post-notify the user, and force a restart as part of the deployment. The upgrade does not complete until the computer is restarted and fails if the user shuts down the computer while it is running. As a result, it is necessary to inform the user that a Windows upgrade is beginning and subsequently that their computer must be restarted.

Deploy Cleanup

After you complete the Phase 3 deployment or run the Phase 1 deployment again, if necessary, use the **Windows 10 Cleanup** software package to remove artifacts and enable future usage of Windows 10 upgrade software packages.

1. From the Deploy menu, go to **Software**.
2. Select **Windows 10 Upgrade Cleanup** and click **Deploy Package**.
3. In the **Targeting** section, select **Set Targeting Criteria** for endpoints to remove Windows 10 upgrade artifacts. For example, you can enter the following targeting filter to clean up endpoints that completed the 21H1 x64 upgrade:

```
Deploy - Installed Software Packages matches .*InPlace Upgrade to Windows 10  
Version 21H1 x64 - Phase3 - Upgrade.*Installed
```



If necessary, replace the software package name to match the package you deployed.

4. Select the desired options according to your environment, click **Save**, and then click **Deploy Software**.

For more information about deployment settings, see [Deploying software on page 55](#).

Troubleshooting

You can access detailed logs for each phase of the Windows Upgrade.

To troubleshoot Windows Setup errors in each phase, review the appropriate log file from an affected endpoint:

- Phase 1 media caching: %temp%\Win10IPU_PreCache.txt
- Phases 1 and 2 compatibility scan: %temp%\Win10IPU_CompatScan.txt
- Phase 3 Windows setup: %temp%\Win10IPU_Upgrade.txt

For problems with Windows Setup, see the appropriate Microsoft log files as described in [Microsoft Support: Log files that are created when you upgrade to a new version of Windows](#).

For general Deploy troubleshooting, see [Troubleshooting Deploy on page 68](#).